

Symbols, Constants, Units and Key Equations

EART23001 Atmospheric Physics and Weather

Using this reference. These are the symbols and key equations used most often in the practice sheets. You are not expected to memorise every numerical constant. Values needed for a calculation are normally supplied in the question. Units shown here are the preferred SI units; hPa, km, μm and other convenient atmospheric units are used where appropriate. The equations in the second half of this reference are collected from across all twelve weeks; the exam itself supplies a shorter table of the constants and equations needed for that specific paper.

Symbols, constants and units

State, thermodynamics and vertical structure

Symbol	Meaning	Typical unit/value
p	pressure	Pa; 1 hPa = 100 Pa
p_0	reference pressure (often the surface value, or 1000 hPa in the potential-temperature definition)	Pa or hPa
T	absolute temperature	K
T_0	reference or surface temperature	K
T_d	dew-point temperature	K or $^{\circ}\text{C}$
ρ	air density	kg m^{-3}
ρ_0	reference or surface air density	kg m^{-3}
z	height above a reference level	m or km
g	acceleration due to gravity	9.81 m s^{-2}
R^*	universal gas constant	$8.314 \text{ J mol}^{-1} \text{ K}^{-1}$
R_d	specific gas constant for dry air	$287 \text{ J kg}^{-1} \text{ K}^{-1}$
R_v	specific gas constant for water vapour	$461.5 \text{ J kg}^{-1} \text{ K}^{-1}$
M_d	molar mass of dry air	$28.97 \times 10^{-3} \text{ kg mol}^{-1}$
c_p	specific heat capacity of dry air at constant pressure	$1004 \text{ J kg}^{-1} \text{ K}^{-1}$
H	pressure/density scale height in a simplified exponential atmosphere	m or km
Γ_{env}	environmental lapse rate, $-dT/dz$	K km^{-1}
Γ_d	dry adiabatic lapse rate	$\approx 9.8 \text{ K km}^{-1}$
Γ_s	saturated (moist) adiabatic lapse rate; not a universal constant	K km^{-1}
θ	dry potential temperature	K
κ	R_d/c_p	≈ 0.286

Water vapour and cloud microphysics

Symbol	Meaning	Typical unit/value
e	water-vapour partial pressure	Pa or hPa
e_s	saturation vapour pressure; subscripts w and i denote liquid water and ice	Pa or hPa
RH	relative humidity	%
r_v	water-vapour mixing ratio (mass of vapour / mass of dry air)	kg kg^{-1} or g kg^{-1}
q	specific humidity (mass of vapour / mass of moist air)	kg kg^{-1}
ϵ	ratio R_d/R_v	≈ 0.622
ρ_v	water-vapour density (partial density of vapour in moist air)	kg m^{-3}
M_w	molar mass of water	$0.018 \text{ kg mol}^{-1}$

L_v	latent heat of vaporisation	$\approx 2.5 \times 10^6 \text{ J kg}^{-1}$
z_{LCL}	lifting condensation level height above the surface	m or km
Γ_{dew}	rate at which a rising parcel's dew point falls with height (fixed mixing ratio)	$\approx 1.8 \text{ K km}^{-1}$
N	cloud-droplet number concentration	m^{-3} or cm^{-3}
r	droplet/hydrometeor radius (the smaller of two radii where both appear, e.g. collision-coalescence)	m or μm
$\mathcal{R}$	collector-drop radius in collision-coalescence growth (always the larger of the two radii); $\mathcal{R}_0$ denotes its initial value	m or μm
ρ_w	density of liquid water	1000 kg m^{-3}
LWC	liquid-water content (liquid-water mass per volume of air)	kg m^{-3} or g m^{-3}
M_2	area-related second moment. For identical droplets, $M_2 = Nr^2$	m^{-1} if N is m^{-3}
M_3	volume/mass-related third moment. For identical droplets, $M_3 = Nr^3$	dimensionless in this simplified form
β_{ext}	volume extinction coefficient; for cloud droplets it is proportional to an area measure such as M_2	m^{-1}
τ	optical depth, approximately $\beta_{ext}\Delta z$ for a uniform layer	dimensionless
A_c	cloud albedo (fraction of incident shortwave radiation reflected)	dimensionless
σ_w	surface tension of liquid water	$\approx 0.075 \text{ N m}^{-1}$ near 10°C
η	dynamic viscosity of air	$\approx 1.7 \times 10^{-5} \text{ Pa s}$
v_t	terminal fall speed	m s^{-1}
E	collection efficiency used in collision-coalescence calculations	dimensionless, $0 \leq E \leq 1$

Atmospheric dynamics

Symbol	Meaning	Typical unit/value
u, v, w	eastward, northward and vertical wind components	m s^{-1}
V	wind-speed magnitude when a component is not required	m s^{-1}
a_p	horizontal pressure-gradient acceleration	m s^{-2}
Ω	Earth's angular rotation rate	$7.292 \times 10^{-5} \text{ s}^{-1}$
ϕ	latitude	degrees or radians
f	Coriolis parameter, $2\Omega \sin \phi$	s^{-1}
v_g	geostrophic wind speed (or a component of the geostrophic wind)	m s^{-1}
$\Delta p / \Delta x$	finite-difference horizontal pressure gradient	Pa m^{-1}
R_c	radius of curvature of curved (gradient-wind) flow	m or km
ζ	relative vorticity; $\zeta + f$ is absolute vorticity	s^{-1}

Radiation and planetary energy balance

Symbol	Meaning	Typical unit/value
$L_\odot$	solar luminosity	$\approx 3.83 \times 10^{26} \text{ W}$
d	distance from the Sun in inverse-square calculations	m
S_0	solar irradiance (solar constant) at mean Earth-Sun distance	$\approx 1361 \text{ W m}^{-2}$
F_{abs}	absorbed radiative flux	W m^{-2}
α	planetary/surface albedo unless a cloud-specific A_c is used	dimensionless
σ_{SB}	Stefan-Boltzmann constant	$5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
T_e	effective radiating temperature	K
T_s	surface temperature in an energy-balance model	K
T_a	atmospheric-layer temperature in a simple greenhouse model	K

ϵ_{eff}	effective grey-atmosphere emissivity relating T_s to outgoing flux; <i>not</i> the same quantity as $\epsilon = R_d/R_v$ above	dimensionless, $\epsilon_{eff} \leq 1$
λ	wavelength	m, μm or nm
λ_{max}	wavelength of maximum blackbody emission	m or μm
b	Wien displacement constant	$2.898 \times 10^{-3} \text{ m K}$

Notation note. r_v is reserved for water-vapour mixing ratio, while r is used for particle radius. σ_{SB} denotes the Stefan–Boltzmann constant and σ_w denotes water surface tension. $\epsilon = R_d/R_v \approx 0.622$ (a moisture ratio) is unrelated to ϵ_{eff} (a radiative emissivity) – the subscript always distinguishes them. Plain R is never used alone in this course: the universal/specific gas constants are always written R^* , R_d or R_v ; a radius of curvature in curved flow is written R_c ; and a collector-drop radius in collision–coalescence is written $\mathcal{R}$, to keep it visually distinct from the smaller droplet radius r it collects. This avoids the notation clashes that are common in atmospheric physics.

Key equations

Using this reference. These are the equations that do the actual computational work in the practice sheets, collected from across all twelve weeks and grouped to match the symbol tables above. A handful – the ideal gas law, the hydrostatic equation, potential temperature, geostrophic balance, and the Stefan–Boltzmann radiative-equilibrium relation – reappear in several different weeks and are worth knowing well before the exam. Values of R_d , c_p , g etc. needed for a calculation are normally supplied in the question; see the symbol tables above for typical values.

State, thermodynamics and vertical structure

Equation	Notes
$p = \rho R_d T$	ideal gas law (dry air)
$R_d = R^*/M_d$	specific gas constant for dry air, from the universal gas constant and dry-air molar mass
$e = \rho_v R_v T$	ideal gas law applied to water vapour alone
$\frac{dp}{dz} = -\rho g$	hydrostatic equation
$\Delta z = \frac{R_d \bar{T}}{g} \ln\left(\frac{p_1}{p_2}\right)$	hypsothetic equation: thickness between two pressure surfaces, using the mean layer temperature $\bar{T}$. This general form is what the exam names; the weekly sheets mostly use the single-temperature special case below
$\rho = \rho_0 \exp(-z/H), \quad p = p_0 \exp(-z/H)$	isothermal exponential atmosphere
$H = \frac{R_d T}{g} = \frac{R^* T}{M g}$	pressure/density scale height
$z = -H \ln\left(\frac{p}{p_0}\right)$	height in terms of pressure, isothermal atmosphere
$p(z) = p_0 \left(\frac{T(z)}{T_0}\right)^{g/(R_d \Gamma)}$	hydrostatic pressure profile for a constant lapse rate Γ , $T(z) = T_0 - \Gamma z$
$\theta = T \left(\frac{1000 \text{ hPa}}{p}\right)^{R_d/c_p}$	potential temperature (Poisson's equation), $R_d/c_p \approx 0.286$
$dU = \delta Q - p dV$	first law of thermodynamics
$Q = m c_p \Delta T$	sensible heat
$Q = m L_v$	latent heat
$p = \sum_i p_i = p_d + e$	Dalton's law of partial pressures; moist air is dry air plus water vapour

Water vapour and cloud microphysics

Equation	Notes
$RH = 100 \frac{e}{e_s(T)}$	relative humidity (an RH_i variant uses e_{si} , saturation over ice)
$r_v = \epsilon \frac{e}{p - e}$	mixing ratio from vapour pressure
$q = \frac{r_v}{1 + r_v}$	specific humidity from mixing ratio
$z_{LCL} \approx 125(T - T_{d0}) \text{ m, with } T, T_d \text{ in } ^\circ\text{C}$	lifting condensation level, quick estimate
$z_{LCL} = \frac{T_0 - T_{d0}}{\Gamma_d - \Gamma_{dew}}$	LCL height, derived from where the parcel's temperature and dew-point lines cross
$\mathcal{S}_{eq}(r) = \exp\left(\frac{A_K}{r}\right), \quad A_K = \frac{2\sigma_w M_w}{\rho_w R^* T}$	Kelvin effect: equilibrium saturation ratio over a curved droplet surface

$N_A r_A^3 = N_B r_B^3$	fixed liquid-water content: $r \propto N^{-1/3}$ between two droplet populations
$M_2 = N r^2, \quad M_3 = N r^3$	area- and volume-related droplet-population moments
$r^2 - r_0^2 = K t$	diffusional (condensational) droplet growth
$v_t = \frac{2(\rho_w - \rho_a) g r^2}{9\eta}$	Stokes terminal fall speed
$\frac{d\mathcal{R}}{dt} = \frac{a N c}{4} \mathcal{R}^b$	collision-coalescence collector-growth rate, $c = \frac{4}{3} \pi r^3$
$\mathcal{R}(t) = \left[\mathcal{R}_0^{1/2} + \frac{a N c}{8} t \right]^2$	solution for $b = 1/2$: two collectors with different $\mathcal{R}_0$ diverge in radius over time
$A_c \approx \frac{X}{X + 15.4}, \quad X = \pi M_2 \Delta z$	approximate cloud-albedo relation from the M_2 moment and cloud depth Δz

Atmospheric dynamics

Equation	Notes
$a_p = \frac{1}{\rho} \left \frac{\Delta p}{\Delta x} \right $	horizontal pressure-gradient acceleration (magnitude)
$f = 2\Omega \sin \phi$	Coriolis parameter
$v_g = a_p / f$	geostrophic wind balance (straight, steady flow)
$\frac{v^2}{R_c} + f v = a_p$	gradient-wind balance for flow curving with radius R_c
$\Delta u_g \approx -\frac{R_d}{f} \frac{\partial T}{\partial y} \ln \left(\frac{p_2}{p_1} \right)$	thermal wind: change in geostrophic wind with height from a horizontal temperature gradient
$\zeta + f \approx \text{constant}$	approximate conservation of absolute vorticity (qualitative Rossby-wave reasoning)

Radiation and planetary energy balance

Equation	Notes
$S_0 = \frac{L_\odot}{4\pi d^2}$	inverse-square law: solar irradiance at distance d from the Sun
$\langle F_{SW} \rangle = \frac{S_0}{4}$	global-mean absorbed shortwave flux (disk-to-sphere averaging), before albedo
$\lambda_{max} T = b$	Wien's displacement law
$\frac{(1 - \alpha) S_0}{4} = \sigma_{SB} T_e^4$	zero-dimensional radiative equilibrium: effective radiating temperature T_e
$F_{out} = \epsilon_{eff} \sigma_{SB} T_s^4$	grey-atmosphere effective emissivity linking surface temperature to outgoing flux
$\sigma_{SB} T_s^4 = 2 F_{abs}$, i.e. $T_s = 2^{1/4} T_e$	one-layer greenhouse model (single absorbing/emitting atmospheric layer)
net flux = absorbed SW – OLR	TOA radiative imbalance by latitude; the driver of poleward heat transport